SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Performance Testing

Date	15 March 2026
Team ID	
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	/submit endpoint (Loan Approval Prediction API)
Test Environment	Local (Flask development server)
Test Date	20 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load test on /submit prediction endpoint	50	60	All requests return prediction within 2 sec
2	Stress test on /submit endpoint	200	120	System remains stable up to 150 VUs, degrades gracefully beyond
3	Spike test - sudden traffic surge	300	30	No crash; error rate stays below 1%
4	Soak test - sustained load	50	1800	Stable CPU/memory usage over time, no leaks

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Well within threshold
2	Response Time (Max)	< 5 seconds	3.8 sec	Pass	Slight increase under peak load
3	Throughput (Req/sec)	> 50 req/sec	62 req/sec	Pass	Handled sustained load smoothly
4	Error Rate	< 1%	0.4%	Pass	Minor errors during spike test
5	CPU Utilization	< 80%	68%	Pass	Stable under peak load
6	Memory Utilization	< 80%	71%	Pass	No memory leaks observed

Step 4: Observations & Analysis

Key Findings:

- The /submit prediction endpoint maintained response times under 2 seconds for up to 150 concurrent users, with throughput scaling linearly.

Bottlenecks Identified:

- Response times increased under heavy concurrent load due to the single-threaded Flask development server used for testing.

Optimization Steps Taken:

- Deployed with Gunicorn using multiple worker processes and enabled response caching for repeated identical requests.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

JMeter Summary Report

Label	# Samples	Average	Min	Max	Error %	Throughput
P Request (/sul	6000	1.2 s	0.4 s	3.8 s	0.4%	62.4/sec